

Latin Square Designs

Blocking on Two Nuisance Sources Simultaneously

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Motivation: Two Sources of Nuisance Variation

From One Blocking Variable to Two

Recall the **Randomized Complete Block Design** (RCBD): we control *one* nuisance source by grouping experimental units into blocks.

In many field and industrial experiments there are **two** distinct nuisance sources acting at the same time:

- ▶ A field with a *north–south* gradient in sunlight *and* an *east–west* gradient in soil moisture.
- ▶ An emissions test where both the *driver* and the *car* influence the measured response.
- ▶ A tasting panel where both the *judge* and the *serving order* affect scores.

Key idea

A **Latin square** lets us block on *two* nuisance factors at once, using only g^2 experimental units for g treatments.

Example: Comparing Five Fertilizers

Goal: compare fertilizers **A, B, C, D, E** on a farm plot with

- ▶ a north–south variation in sunlight (rows), and
- ▶ an east–west variation in soil humidity (columns).

Row	Column (dry $\longleftrightarrow$ humid)				
	1	2	3	4	5
I (less sun)	A	B	C	D	E
II	C	D	E	A	B
III	E	A	B	C	D
IV	B	C	D	E	A
V (more sun)	D	E	A	B	C

Each treatment appears **exactly once in every row** and **exactly once in every column**. This arrangement is a **Latin square**.

What Is a Latin Square?

Definition (Latin square of order g)

A **Latin square** is a $g \times g$ array filled with g symbols (the treatments) such that each symbol occurs **exactly once in each row** and **exactly once in each column**.

Consequences for design:

Double blocking Rows form one blocking variable, columns the other.

Economy Only g^2 runs are needed, versus g^3 for a full three-factor crossing of treatment $\times$ row $\times$ column.

Balance Every treatment is compared against the same set of row and column conditions, so treatment contrasts are *orthogonal* to both blocking factors.

Remark

The price of the economy is a strong *additivity* assumption: rows, columns, and treatments do not interact.

A Worked Example: Automobile Emissions

The Automobile Emissions Study

An engineer studies four fuel additives designed to reduce pollution.

Treatments Additives **A, B, C, D** ($i = 1, 2, 3, 4$)

Row block Drivers I, II, III, IV ($j = 1, 2, 3, 4$)

Column block Cars 1, 2, 3, 4 ($k = 1, 2, 3, 4$)

Unit each (driver, car) combination is one test drive

Response emission-reduction index for that drive

Both the driver and the car are nuisance sources we want to remove. With four treatments, a Latin square of order 4 needs only $4^2 = 16$ test drives.

Driver	Cars				Driver
	1	2	3	4	avg
I	A:19	B:24	D:23	C:26	23
II	D:23	C:24	A:19	B:30	24
III	B:15	D:14	C:15	A:16	15
IV	C:19	A:18	B:19	D:16	18
Car avg	19	20	19	22	20

Treatment means: A: 18 B: 22 C: 21 D: 19

Grand mean $\bar{y}_{\bullet\bullet\bullet} = 20$. Notice the strong spread across drivers (15 to 24) — driver looks like an important nuisance source.

The Data in Long Format

driver	car	trt	reduct	driver	car	trt	reduct
1	1	A	19	3	1	B	15
1	2	B	24	3	2	D	14
1	3	D	23	3	3	C	15
1	4	C	26	3	4	A	16
2	1	D	23	4	1	C	19
2	2	C	24	4	2	A	18
2	3	A	19	4	3	B	19
2	4	B	30	4	4	D	16

Three factors — *driver*, *car*, *trt* — plus the response *reduct*. Each factor has four balanced levels.

Reading the Design: Questions to Keep in Mind

Before any formal analysis, look at the treatment means *within* each blocking factor:

- ▶ Are there **driver effects**? (Driver averages: 23, 24, 15, 18)
- ▶ Are there **car effects**? (Car averages: 19, 20, 19, 22)
- ▶ **Which block effect is stronger?**

Remark

Driver averages vary a lot (a 9-unit spread); car averages barely move (a 3-unit spread). We expect *driver* to explain much more variation than *car*. The ANOVA will confirm this.

The Latin Square Model

Latin square model

$$y_{ijk} = \mu + \underbrace{\alpha_i}_{\text{treatment}} + \underbrace{\beta_j}_{\text{row}} + \underbrace{\gamma_k}_{\text{column}} + \varepsilon_{ijk},$$

with the usual constraints and error assumption

$$\sum_i \alpha_i = \sum_j \beta_j = \sum_k \gamma_k = 0, \quad \varepsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

- ▶ There are g^2 units, but the three indices are *not* free: once the row j and column k are fixed, the design forces a single treatment $i = i(j, k)$.
- ▶ The model is **additive** — no treatment $\times$ row, treatment $\times$ column, or row $\times$ column interaction terms.

Parameter Estimates

Because the design is balanced, the least-squares estimates take the familiar one-way form:

$$\begin{aligned}\hat{\mu} &= \bar{y}_{\bullet\bullet\bullet}, & \hat{\alpha}_i &= \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet} \quad (\text{treatment}), \\ \hat{\beta}_j &= \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet} \quad (\text{row}), & \hat{\gamma}_k &= \bar{y}_{\bullet\bullet k} - \bar{y}_{\bullet\bullet\bullet} \quad (\text{column}).\end{aligned}$$

- ▶ Every effect is a simple deviation of a marginal mean from the grand mean.
- ▶ Orthogonality (from the balance) means the three sets of estimates do not depend on one another — dropping columns would not change $\hat{\alpha}_i$.

Remark

All estimation and testing can be done with the same `lm()` / `anova()` workflow you already use for two-way ANOVA.

How Do Latin-Square Designs Work?

Question. What is the expected value of the treatment-A mean $\bar{y}_{i\bullet\bullet}$ for $i = A$?

In the emissions square, treatment A appears in four cells:

$$\bar{y}_{i\bullet\bullet}^{(A)} = \frac{1}{4}(y_{A11} + y_{A32} + y_{A43} + y_{A24}).$$

Using $E(y_{ijk}) = \mu + \alpha_i + \beta_j + \gamma_k$,

$$\begin{aligned} E[\bar{y}_{i\bullet\bullet}^{(A)}] &= \frac{1}{4} \left[(\mu + \alpha_A + \beta_1 + \gamma_1) + (\mu + \alpha_A + \beta_3 + \gamma_2) \right. \\ &\quad \left. + (\mu + \alpha_A + \beta_4 + \gamma_3) + (\mu + \alpha_A + \beta_2 + \gamma_4) \right] \\ &= \mu + \alpha_A + \underbrace{\frac{1}{4} \sum_j \beta_j}_{=0} + \underbrace{\frac{1}{4} \sum_k \gamma_k}_{=0} = \mu + \alpha_A. \end{aligned}$$

Why the balance matters

Because A meets *every* row and *every* column exactly once, the row and column effects **cancel** — the treatment mean is a clean estimate of $\mu + \alpha_A$.

The ANOVA Table

ANOVA Table for a Single Latin Square

Source	df	SS	MS	F
Row block	$g - 1$	SS_{row}	$SS_{\text{row}}/(g - 1)$	MS_{row}/MS_E
Column block	$g - 1$	SS_{col}	$SS_{\text{col}}/(g - 1)$	MS_{col}/MS_E
Treatment	$g - 1$	SS_{trt}	$SS_{\text{trt}}/(g - 1)$	MS_{trt}/MS_E
Error	$(g - 2)(g - 1)$	SS_E	$SS_E/[(g - 2)(g - 1)]$	
Total	$g^2 - 1$	SS_T		

Note: error degrees of freedom shrink fast

With three sets of $g - 1$ effects removed, only $(g - 2)(g - 1)$ df remain for error. For $g = 3$ that is just **2** df — Latin squares of order 3 or 4 give very few error df, so replication is often needed.

The Sums of Squares

Each blocking / treatment SS is a marginal sum of squared effects, scaled by g :

$$SS_{\text{row}} = g \sum_j \hat{\beta}_j^2 = g \sum_j (\bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet \bullet \bullet})^2,$$

$$SS_{\text{col}} = g \sum_k \hat{\gamma}_k^2 = g \sum_k (\bar{y}_{\bullet \bullet k} - \bar{y}_{\bullet \bullet \bullet})^2,$$

$$SS_{\text{trt}} = g \sum_i \hat{\alpha}_i^2 = g \sum_i (\bar{y}_{i \bullet \bullet} - \bar{y}_{\bullet \bullet \bullet})^2.$$

The error and total sums of squares complete the decomposition:

$$SS_E = \sum_{ijk} (y_{ijk} - \bar{y}_{i \bullet \bullet} - \bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet \bullet k} + 2\bar{y}_{\bullet \bullet \bullet})^2 = SS_T - SS_{\text{row}} - SS_{\text{col}} - SS_{\text{trt}},$$

$$SS_T = \sum_{ijk} (y_{ijk} - \bar{y}_{\bullet \bullet \bullet})^2.$$

Remark

$SS_T = SS_{\text{row}} + SS_{\text{col}} + SS_{\text{trt}} + SS_E$: total variation splits cleanly into three explained pieces plus residual — a consequence of orthogonality.

ANOVA in R

Fit the additive three-factor model and read the ANOVA table:

```
emis ← read.table("emission.txt", header = TRUE)

lm1 ← lm(reduct ~ factor(driver) + factor(car) + factor(trt),
         data = emis)

anova(lm1)
```

Output:

```
Response: reduct
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
factor(driver)	3	216.0	72.00	13.5	0.00447	**
factor(car)	3	24.0	8.00	1.5	0.30717	
factor(trt)	3	40.0	13.33	2.5	0.15649	
Residuals	6	32.0	5.33			

Interpreting the Emissions ANOVA

Source	F	p-value	Conclusion
Driver (row)	13.5	0.0045	strong effect
Car (column)	1.5	0.307	no evidence
Treatment	2.5	0.156	not significant

- ▶ **Driver** is by far the dominant nuisance source — exactly what the marginal means suggested.
- ▶ **Car** contributes little; blocking on it "cost" 3 df for almost no gain here, but that is only clear *after* the fact.
- ▶ With just 6 error df, the treatment test has **low power**; a real additive difference could easily go undetected.

Efficiency, Assumptions, and Practical Notes

Was Double Blocking Worth It?

Relative efficiency (RE) compares the Latin square (LS) to a completely randomized design (CRD) that ignores both blocks:

$$\widehat{\text{RE}}_{\text{LS:CRD}} = \frac{\text{MS}_{\text{row}} + \text{MS}_{\text{col}} + (g - 1) \text{MS}_E}{(g + 1) \text{MS}_E}.$$

For the emissions data ($\text{MS}_{\text{row}} = 72$, $\text{MS}_{\text{col}} = 8$, $\text{MS}_E = 5.33$, $g = 4$):

$$\widehat{\text{RE}} = \frac{72 + 8 + 3(5.33)}{5(5.33)} \approx 3.6.$$

Reading the number

A CRD would need roughly **3.6 times** as many runs to match the precision of this Latin square. Almost all of the gain comes from blocking on *driver*; column (car) blocking added little.

Assumptions and How to Check Them

The Latin square inherits the usual linear-model assumptions, plus one that is specific to the design:

Additivity rows, columns, and treatments do not interact — *the critical extra assumption*.

Only $(g - 2)(g - 1)$ df remain, so there is no way to fit interactions.

Normal errors check with a Q–Q plot of residuals.

Constant variance check residuals vs. fitted values.

Independence guaranteed by proper randomization of the square.

Caution: hidden interaction

If additivity fails (e.g. an additive works well only for one driver), that interaction is confounded with error, inflating MS_E and *deflating* the treatment F. A Tukey one-degree-of-freedom test for non-additivity can help diagnose this.

Diagnostics and Follow-up in R

Residual checks and multiple comparisons after fitting `lm1`:

```
# Assumption checks
par(mfrow = c(1, 2))
plot(lm1, which = 1)    # residuals vs fitted
plot(lm1, which = 2)    # Normal Q-Q

# Pairwise treatment comparisons (only if trt were significant)
aov1 <- aov(reduct ~ factor(driver) + factor(car) + factor(trt),
            data = emis)
TukeyHSD(aov1, "factor(trt)")
```

Remark

Here the treatment F is not significant ($p = 0.156$), so we would *not* chase pairwise differences; the Tukey step is shown only for the workflow.

Randomizing a Latin Square

A valid Latin square must be *randomly* chosen, not just written down:

1. Start from any standard square of order g .
2. Randomly permute the **rows**.
3. Randomly permute the **columns**.
4. Randomly assign the **treatment labels** to the symbols.

Note

Randomization protects against systematic patterns in the two blocking directions and justifies the additive error model. In R, `agricolae::design.lsd()` builds and randomizes the square for you.

When to Use — and When to Avoid — a Latin Square

Use it when:

- ▶ there are exactly two nuisance sources to block, and
- ▶ the number of row levels = column levels = number of treatments = g , and
- ▶ additivity is plausible.

Be cautious when:

- ▶ g is small (3 or 4) — too few error df; **replicate** the square to recover power.
- ▶ interactions between treatments and blocks are likely.
- ▶ the row, column, and treatment counts cannot all be made equal.

Looking ahead

Replicated Latin squares, Graeco-Latin squares (three blocking directions), and crossover designs all extend this same idea.

Summary

- ▶ A **Latin square** blocks on **two** nuisance factors simultaneously with only g^2 runs.
- ▶ Model is *additive*: $y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \varepsilon_{ijk}$, fitted with ordinary `lm()`.
- ▶ The ANOVA splits total variation into row, column, treatment, and error, each treatment/block SS having $g - 1$ df.
- ▶ Error $df = (g - 2)(g - 1)$ is small — mind the power, and replicate when g is small.
- ▶ In the emissions study: strong **driver** effect, weak car effect, no significant treatment effect, and $RE \approx 3.6$ versus a CRD.